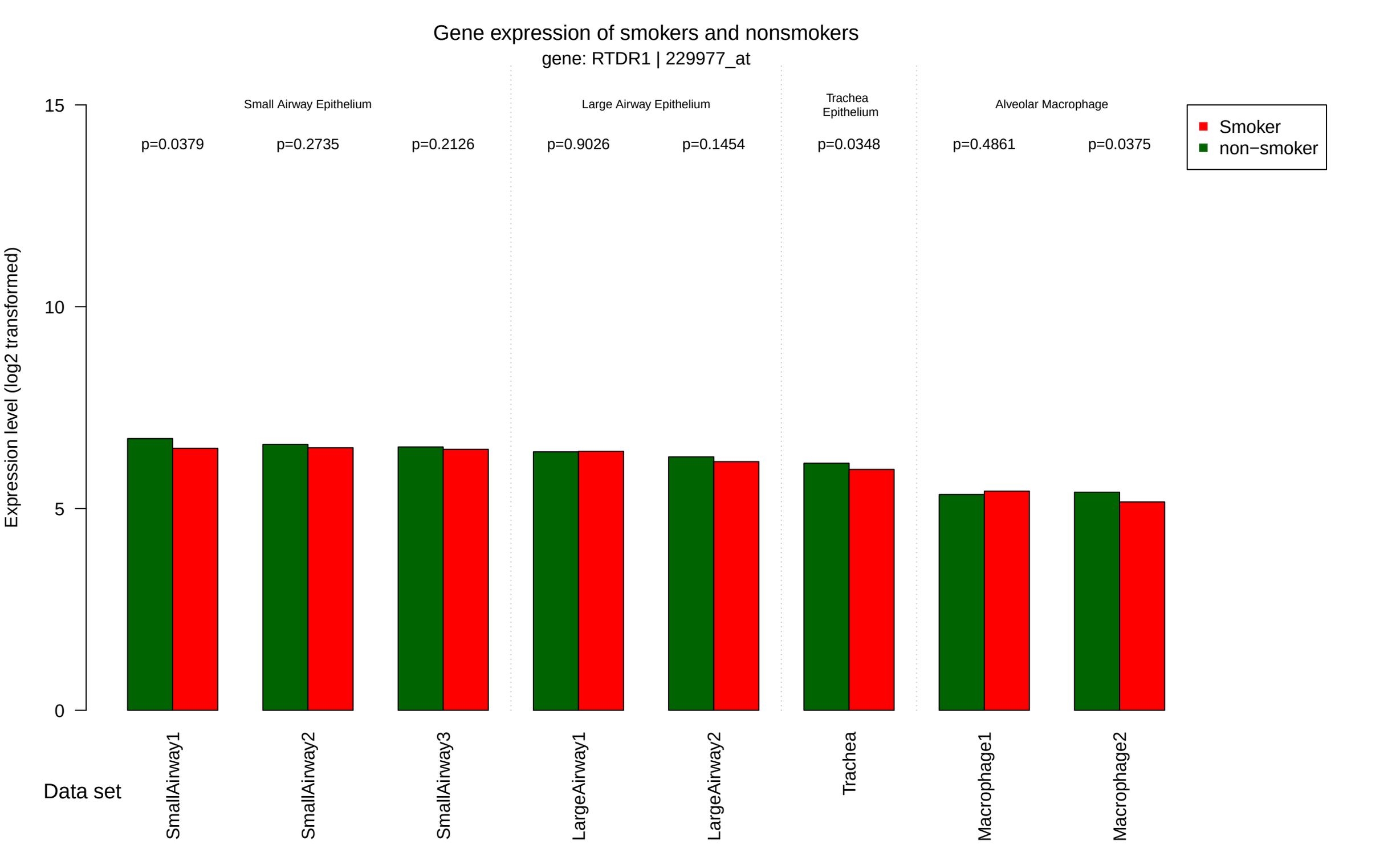
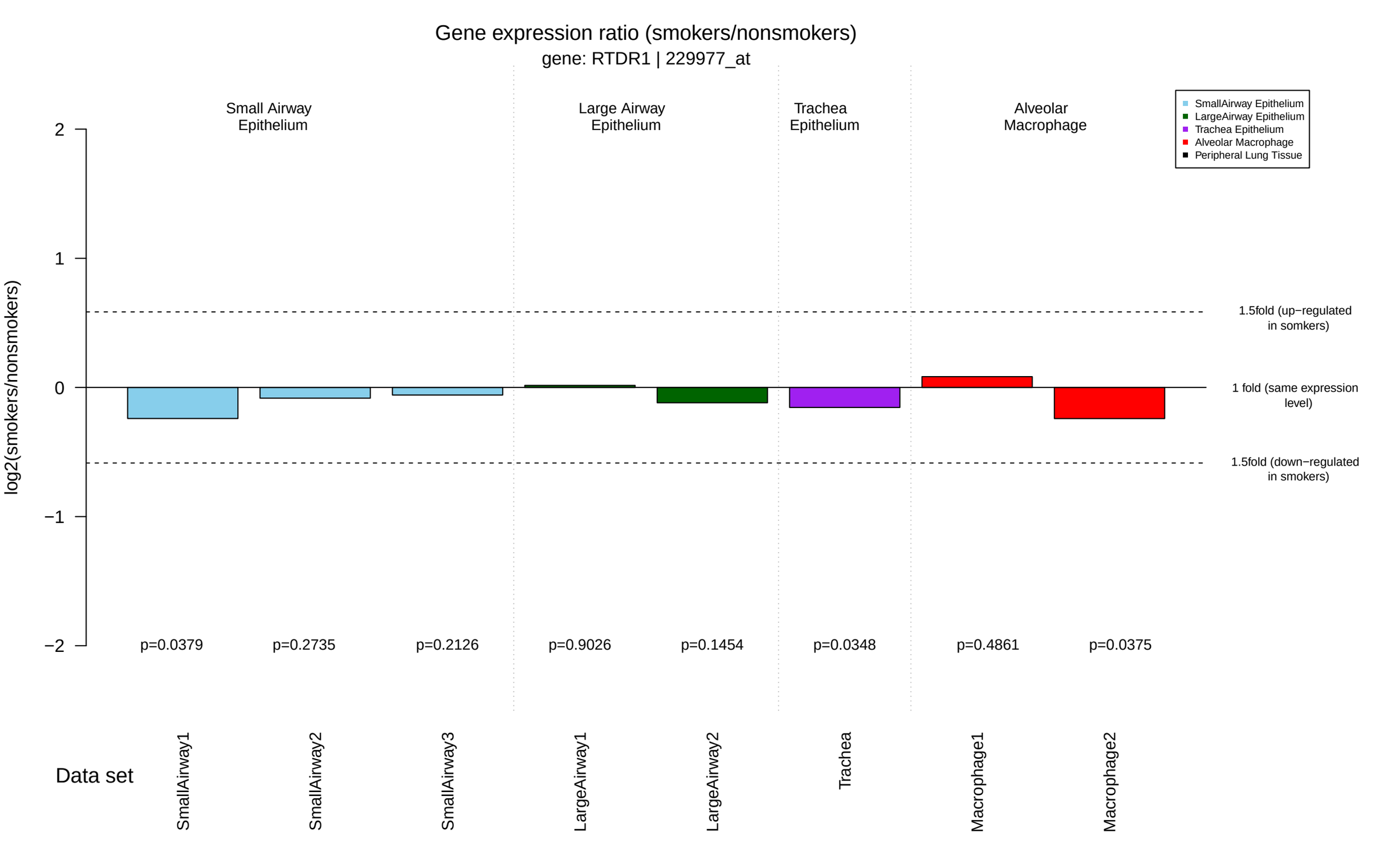
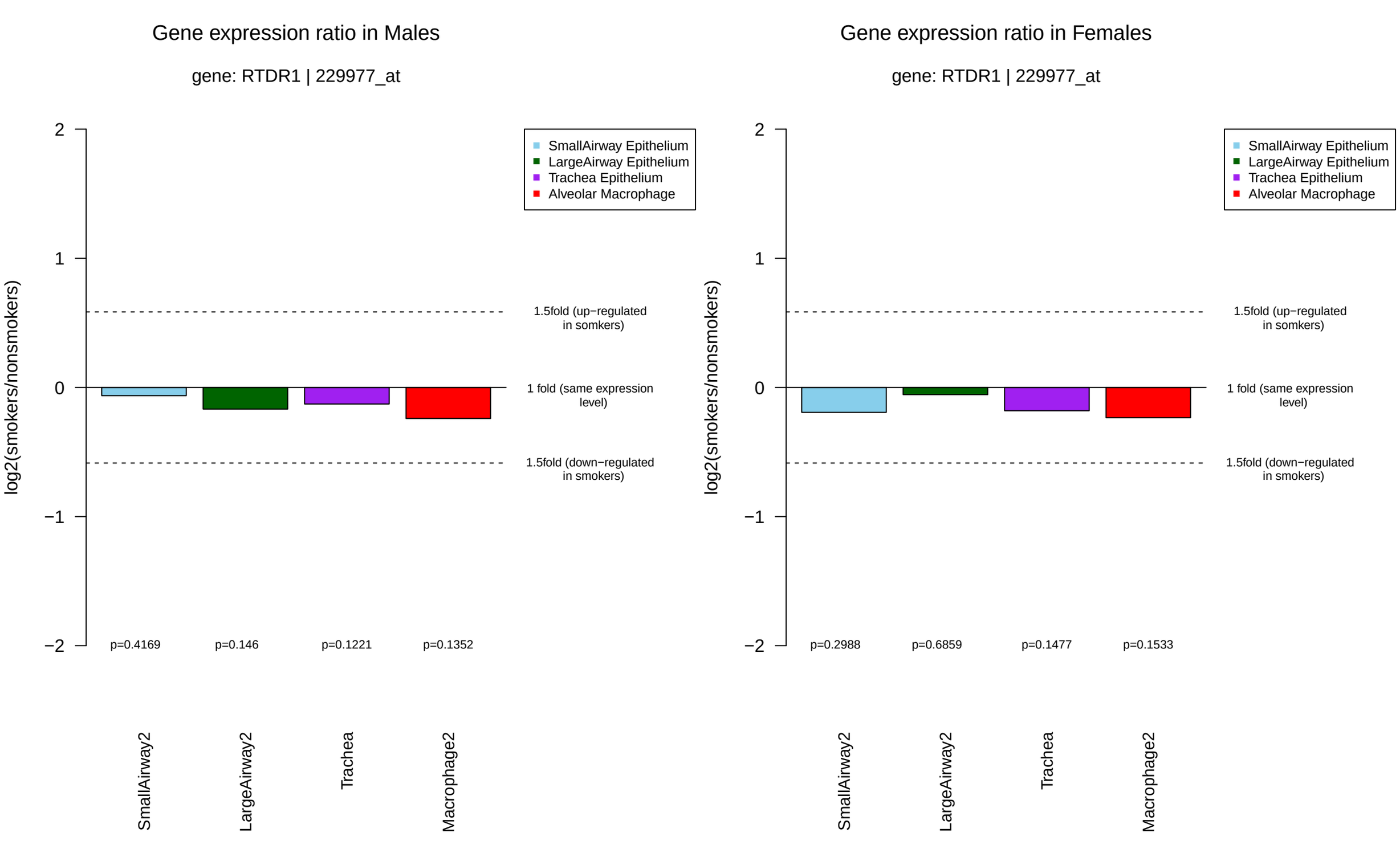
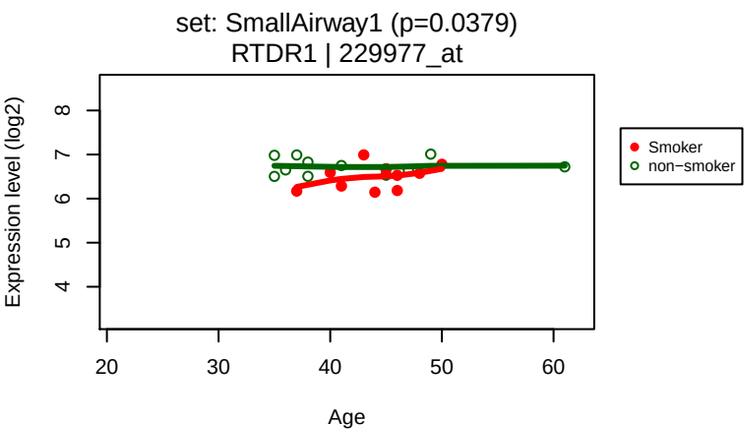
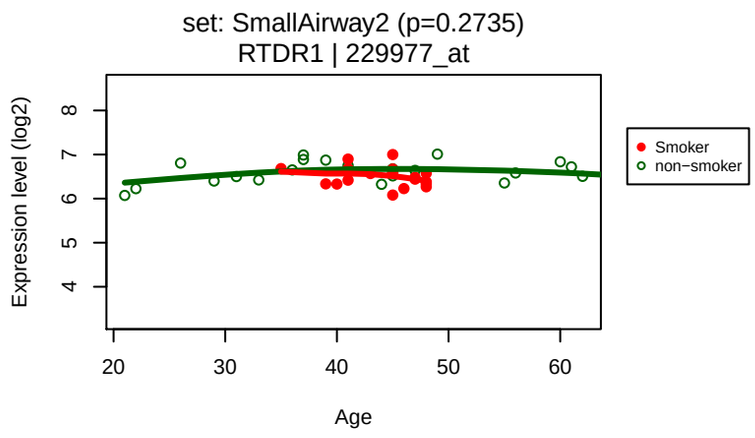
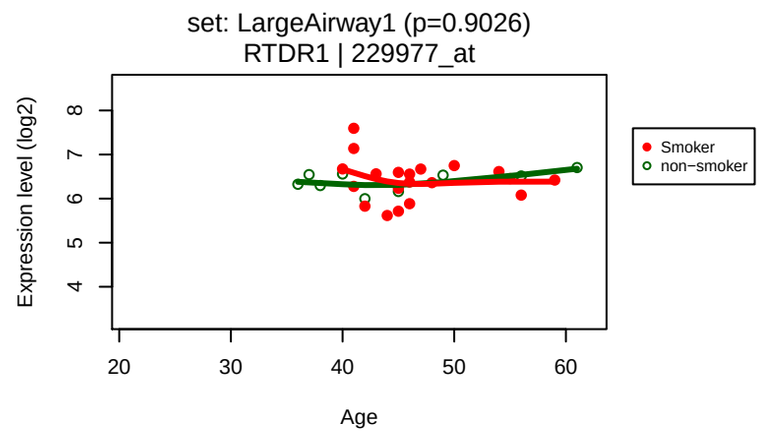

Gene expression of smokers and nonsmokers
gene: RTDR1 | 229977_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0379
p=0.2735
p=0.2126
p=0.9026
p=0.1454
p=0.0348
p=0.4861
p=0.0375
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

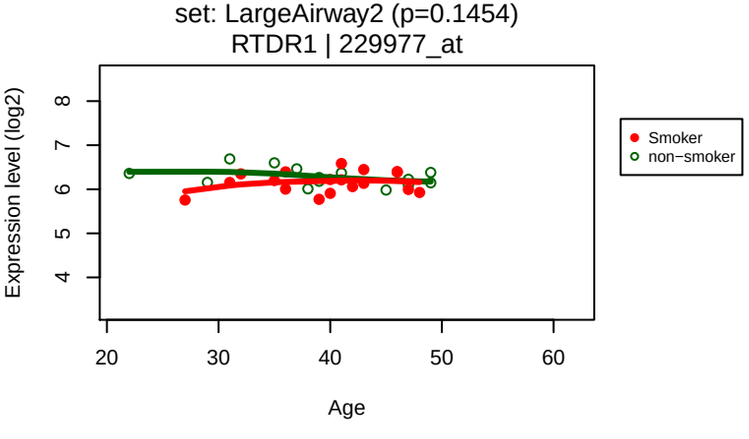

Gene expression ratio (smokers/nonsmokers)
gene: RTDR1 | 229977_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0379
p=0.2735
p=0.2126
p=0.9026
p=0.1454
p=0.0348
p=0.4861
p=0.0375
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

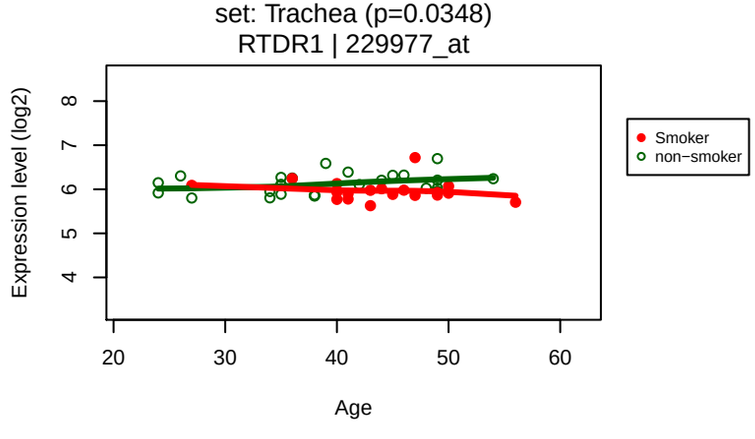

Gene expression ratio in Males
gene: RTDR1 | 229977_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4169
p=0.146
p=0.1221
p=0.1352
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTDR1 | 229977_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2988
p=0.6859
p=0.1477
p=0.1533
SmallAirway2
LargeAirway2
Trachea
Macrophage2

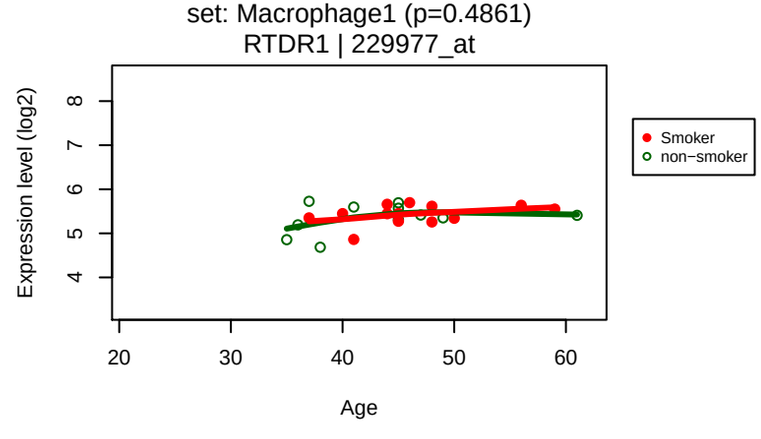

set: SmallAirway1 (p=0.0379)
RTDR1 | 229977_at
Expression level (log2)
Age
Smoker
non−smoker

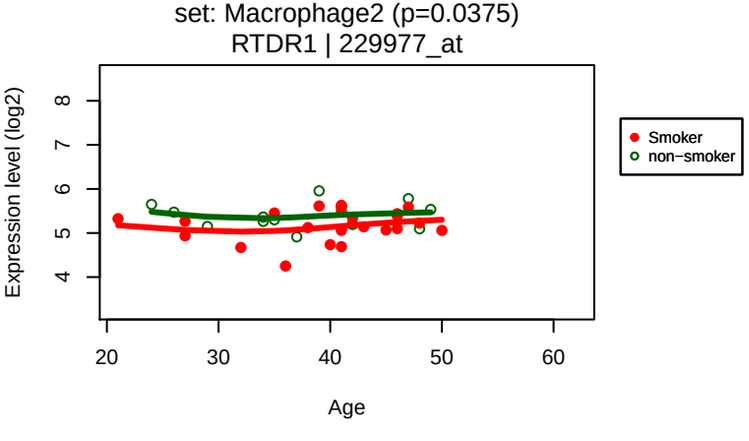

set: SmallAirway2 (p=0.2735)
RTDR1 | 229977_at
Expression level (log2)
Age
Smoker
non−smoker

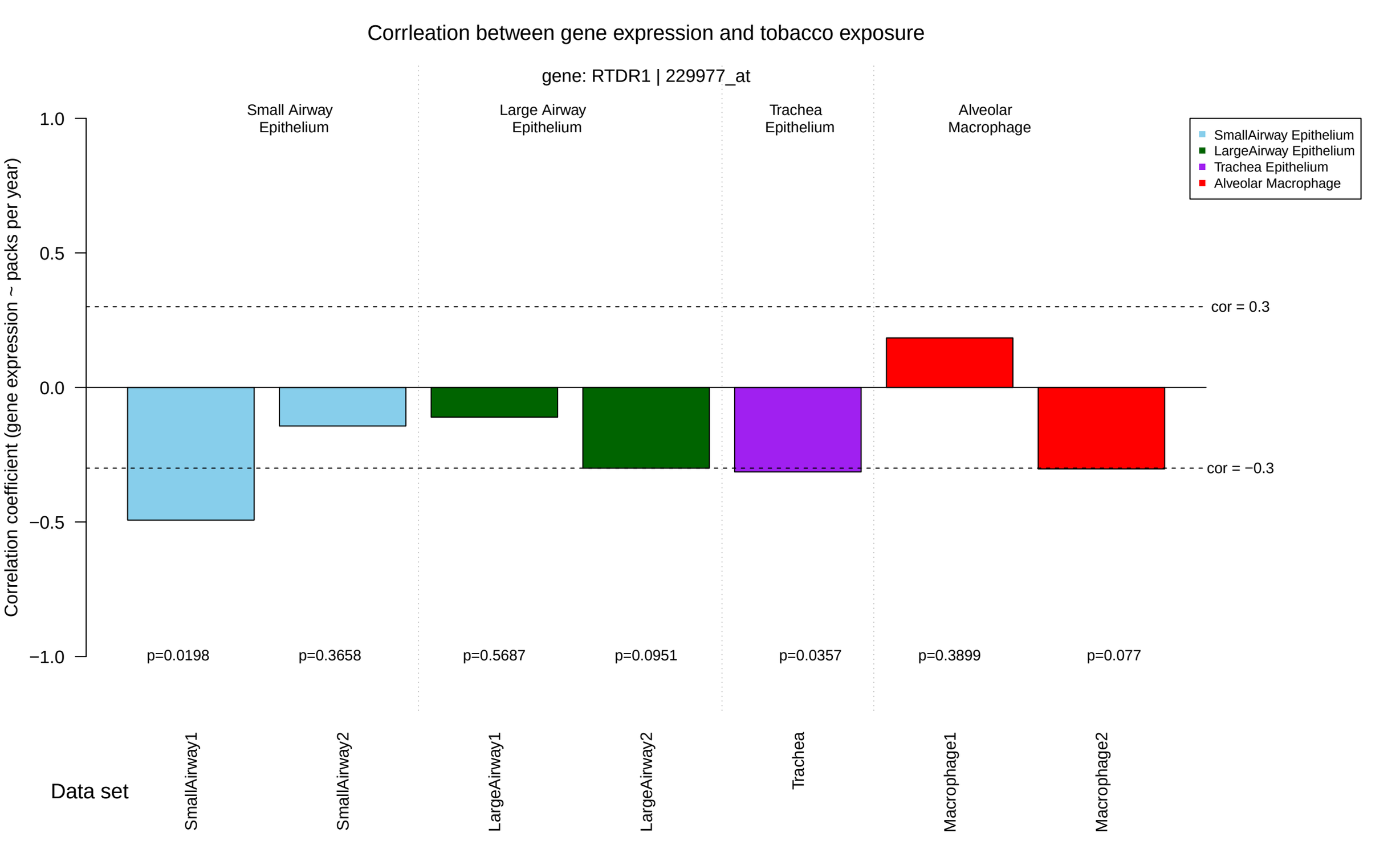

set: LargeAirway1 (p=0.9026)
RTDR1 | 229977_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1454)
RTDR1 | 229977_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0348)
RTDR1 | 229977_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4861)
RTDR1 | 229977_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0375)
RTDR1 | 229977_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTDR1 | 229977_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0198
p=0.3658
p=0.5687
p=0.0951
p=0.0357
p=0.3899
p=0.077
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2